ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# MECHATRONICS

## PAPER 1 · PRACTICE A · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice a.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

## SECTION A

## QUESTION 1

**1.1** 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 A ✓ 1.1.5 C ✓ 1.1.6 C ✓ 1.1.7 A ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

**1.2** 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

**1.3** 1.3.1 F ✓ 1.3.2 I ✓ 1.3.3 D ✓ 1.3.4 G ✓ 1.3.5 A ✓ 1.3.6 H ✓ 1.3.7 C ✓ 1.3.8 B ✓ 1.3.9 J ✓ 1.3.10 E ✓ (10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2

**2.1** A system that combines mechanical parts, electronics (sensors and actuators) and a computer/controller to perform a task automatically ✓✓. Sliding door: input – a motion (PIR / microwave) sensor detects a person ✓; process – the controller decides to open the door and for how long ✓; output – a motor with a belt or rack opens and closes the door ✓; a limit switch / safety sensor gives feedback ✓ (6)

**2.2** Any FOUR: wear safety glasses and safety shoes ✓; no loose clothing, jewellery or long loose hair ✓; switch off and isolate machines before maintenance ✓; keep the work area clean and tidy ✓; use guards on machines; know where the emergency stop and fire extinguisher are; use tools only for their purpose (4)

**2.3** (a) tin snips ✓ (b) spanner / socket wrench ✓ (c) wire strippers ✓ (d) spirit level ✓ (4)

**2.4** First class – fulcrum between effort and load ✓, e.g. seesaw, scissors, crowbar ✓; second class – load between fulcrum and effort ✓, e.g. wheelbarrow, bottle opener ✓; third class – effort between fulcrum and load ✓, e.g. tweezers, fishing rod, human forearm ✓ (6)

**[20]**

## QUESTION 3

**3.1.1** $\text{Load} \times 0{,}3 = 150 \times 1{,}2$ ✓; $\text{load} = 180 \div 0{,}3$ ✓ $= 600$ N ✓ (3)

**3.1.2** $\text{MA} = \text{load} \div \text{effort} = 600 \div 150$ ✓ $= 4$ ✓ (2)

**3.2.1** $\text{GR} = \text{driven} \div \text{driver} = 60 \div 20$ ✓ $= 3:1$ ✓ (2)

**3.2.2** $\text{N} = 300 \div 3$ ✓ $= 100$ r/min ✓ (2)

**3.2.3** The torque increases ✓ – about three times larger (ignoring losses), because speed is reduced ✓. (2)

**3.2.4** An idler gear between them ✓ (1)

**3.3** $\text{Effort} = \text{load} \div \text{number of supporting ropes} \checkmark = 800 \div 4 \checkmark = 200\ \text{N} \checkmark$ (3)

**3.4** $\text{(a) } W = F \times s = 50 \times 1{,}5 \checkmark = 75\ \text{J} \checkmark \text{ (b) } P = W \div t = 75 \div 3 \checkmark = 25\ \text{W} \checkmark$ (4)

**3.5** Any TWO: lubrication (oil, grease) ✓; bearings ✓; smooth surfaces / polishing; correct alignment (2)

**[21]**

## QUESTION 4

**4.1** Battery symbol ✓; switch ✓; fuse ✓; motor (M in a circle) ✓; all in series in a closed circuit ✓ (5)

**4.2** $P = V \times I = 24 \times 2 \checkmark = 48\ \text{W} \checkmark$ (2)

**4.3** Any THREE: air is free and readily available ✓; clean – leaks do not cause oil spills ✓; fast movement ✓; cheaper components; safe in explosive or wet environments; no return lines needed (3)

**4.4** When the button is pressed, the valve opens and compressed air flows into the cylinder ✓, pushing the piston out (outstroke) ✓; when the button is released, the valve exhausts the air to the atmosphere ✓ and the spring pushes the piston back (instroke) ✓. (4)

**[14]**

## QUESTION 5

**5.1** A sensor detects a physical quantity and converts it into an electrical signal (input) ✓; an actuator converts an electrical signal into movement or another physical action (output) ✓. Sensors (any TWO) ✓✓: LDR, thermistor, ultrasonic, limit switch, proximity sensor. Actuators (any TWO) ✓✓: motor, solenoid, servo, pneumatic cylinder, relay (6)

**5.2** (a) inductive proximity sensor ✓ (b) LDR / light sensor ✓ (c) reed switch with magnet (or a micro switch) ✓ (3)

**5.3** Input: light sensor (LDR) ✓ (or real-time clock); controller: microcontroller (e.g. Arduino) / PLC ✓; output: motor or linear actuator to open and close the door ✓ (limit switches may also be named) (3)

**5.4** Moving parts wear and dust or dirt builds up, which can cause breakdowns and unsafe operation ✓. Any TWO tasks ✓✓: lubricate moving parts; check and tighten belts and chains; clean sensors; check wiring and connectors; drain water from the air compressor; replace worn parts (3)

**[15]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**